

Structural Consistency Is Not Retrieval Utility: An Exact-and-Heuristic Audit of Preference-Graph Repair for Multi-Ranker Retrieval

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Abstract

Purpose: Preference-graph repair is often treated as if restoring acyclicity should improve ranking quality. This study asks whether, for score-derived multi-ranker retrieval graphs, heuristic or exact minimum-weight feedback-arc-set repair yields a multiplicity-corrected retrieval gain – or whether structural consistency and retrieval utility must be reported as separate claims.

Methods: On four public benchmarks, query-level preference graphs were built from stored BM25, TF-IDF, and MiniLM scores under three vote-construction regimes. Rankings from unrepaired, greedy-repaired, and exactly repaired graphs were compared with paired sign-flip tests and Holm correction. Exact SCIP solutions served as a diagnostic control on heuristic under-repair, not as a production solver.

Results: Construction dominated structure: a conservative regime was acyclic throughout, while a permissive regime was cyclic for most queries; normalizing scores reduced BM25’s conditional edge-weight share from **0.988** to **0.512**. Repair removed non-trivial contradictory weight and, pooled across vote regimes when the pool exceeded the cutoff, changed top-*k* membership by **10.6%** on average, yet no repaired-versus-unrepaired normalized discounted cumulative gain (nDCG) comparison survived Holm correction in the canonical, larger-pool, or exact-repair families. Graph-free fusion was competitive (CombSUM **0.554** versus **0.546** for the best repaired hybrid). Exact repair removed less weight than greedy repair yet still yielded no corrected retrieval gain.

Conclusion: Under this protocol, preference-graph repair is a real structural intervention but not a validated nDCG optimization step. Improving the repair objective – even to certified optimality – does not substitute for downstream

evaluation; treat acyclicity metrics as structural diagnostics, not as proxies for retrieval utility.

Keywords: preference graphs, feedback arc set, graph repair, rank aggregation, information retrieval, retrieval evaluation

1 Introduction

Pairwise preferences are a natural and increasingly common form of ranking evidence. Rather than committing to a single scalar relevance score, a pairwise judgment states only that one document should be ranked ahead of another for a given query. This form of evidence arises whenever several ranking signals are available for the same query – lexical and dense retrievers, cross-encoders, or large language models used as pointwise, pairwise, or listwise judges – and it arises directly whenever a system elicits pairwise comparisons instead of absolute scores. Aggregating such judgments across several rankers or several repeated comparisons for the same query produces a natural representation: a query-specific weighted directed preference graph, in which an edge $u \rightarrow v$ records support for ranking document u ahead of document v , and the edge weight records how strongly that support should count.

That representation is attractive because it preserves disagreement rather than collapsing it. It is also structurally fragile. Because the graph is built from several partially disagreeing sources, it can contain directed cycles – $u \rightarrow v \rightarrow w \rightarrow u$ – so that no single total order is consistent with every retained edge. Cycles of this kind are not a pathological edge case; they are the expected outcome whenever comparison evidence is aggregated across imperfectly correlated sources. Because a cycle is a case where the evidence contradicts itself, it is natural to treat acyclicity, or a reduction in cyclic and contradictory structure, as a desirable property of the graph, and cycle repair (removing or reversing a minimum-weight set of conflicting edges) as an attractive preprocessing step before a ranking is read off the graph. This intuition is plausible on its face: a graph that no longer contradicts itself seems like a strictly better input to whatever ranking rule is applied to it next.

That plausibility, however, rests on treating four distinct stages of a preference-graph pipeline as if they were one. *Construction* decides which pairwise votes are retained and how they are weighted, and is itself sensitive to score normalization, vote semantics, and thresholding choices made before any graph exists. *Structural repair* operates on the resulting graph and optimizes a purely graph-internal objective – the total weight of edges that must be removed or reversed to restore acyclicity. *Ranking extraction* reads a final ordering off the (possibly repaired) graph using some scoring rule. *Downstream evaluation* then compares that ordering against relevance judgments using a retrieval metric that has no direct relationship to the repair objective. Repair minimizes disagreement with the graph’s own edges; retrieval metrics score agreement with an external notion of relevance. Nothing in the construction of the repair objective guarantees that reducing the former improves the latter, and nothing in the standard presentation of repair work makes that gap explicit.

This gap is the question prior work leaves open: does improving a preference graph’s structural consistency – as measured by the repair objective itself – also improve the retrieval effectiveness of the ranking eventually read off that graph? Existing evidence does not answer this cleanly, for three reasons that compound. First, essentially all empirical evaluations of preference-graph repair use heuristic repair methods, so any observed retrieval effect, positive or null, is open to the objection that it reflects the heuristic’s suboptimality rather than a property of repair itself: a null result could always be dismissed as “the heuristic simply did not repair enough.” Second, graph construction is rarely held explicit and varied alongside repair, so it is difficult to attribute an observed effect (or its absence) specifically to the repair step rather than to upstream choices that determine how much contradictory structure the graph contained in the first place. Third, studies that report both structural diagnostics (cyclicity, removed edge weight) and a retrieval metric do not generally test whether the two move together under corrected statistical inference; an improvement in the former is often reported alongside, and implicitly read as evidence for, the latter.

This paper addresses that gap with an empirical and methodological study, not with a new ranking model or a new optimization algorithm. We hold graph construction explicit across three vote-construction regimes, and we compare unrepaired, greedy-repaired, and exactly repaired preference graphs on the same candidate sets and stored BM25, TF-IDF, and MiniLM scores. Exact minimum-weight feedback-arc-set (MWFAS) repair, solved to certified optimality with an open-source mixed-integer solver, is included as a diagnostic control: if an optimal repair still fails to produce a reliable retrieval gain of the scale detectable under the protocol, greedy under-repair cannot explain a null of that magnitude. The central empirical pattern is stronger than the observation that repair acts only on cyclic graphs. Construction choices determine whether repairable inconsistency exists at all; when the candidate pool exceeds the evaluation cutoff, repair changes top- k membership by 10.6% on average across the three vote-construction regimes; yet no repaired-versus-unrepaired normalized discounted cumulative gain (nDCG) comparison survives Holm-corrected paired testing in the canonical, larger-pool, or exact-repair families. Simple graph-free fusion remains competitive with the best repaired graph method, reinforcing that structural consistency and retrieval utility are separate claims (Section 6).

The contribution is therefore threefold:

1. An empirical separation result: in score-derived multi-ranker retrieval, graph-construction choices determine whether contradictory structure exists to repair, and repair can alter which documents enter the evaluated top- k without producing a Holm-surviving nDCG gain.
2. A diagnostic exact-repair control: under matched pools and extraction rules, comparing unrepaired, greedy-repaired, and SCIP-optimal MWFAS-repaired graphs rules out greedy under-repair as an explanation for a retrieval gain of the scale detectable under the evaluated protocol in the reported comparison families.
3. A corrected multi-family null against graph-free anchors: no repaired-versus-unrepaired nDCG family survives Holm correction, while CombSUM and reciprocal

rank fusion remain competitive with repaired graph methods, so structural consistency is not shown to be a reliable surrogate for retrieval utility under this protocol.

Scope of evidence.

The confirmatory evidence is score-derived multi-ranker retrieval on four public benchmarks (SciDocs, FiQA, HotpotQA, and BRIGHT) with three base scorers and three vote-construction regimes. Confirmatory retrieval inference is query-level within reported Holm families; the 1,025 certified exact solves (of 1,026 nominal query-regime combinations; Section 5.4) establish solver coverage on the studied graphs, not 1,025 independent retrieval effects. A six-query real-large-language-model pilot is directional only and is not treated as confirmation. The results do not establish behavior for cross-encoder or large-scale LLM rerankers, learned fusion, or interactive retrieval, and they do not prove that repair never helps outside this protocol.

2 Related Work

2.1 Pairwise Ranking and Preference Aggregation

Representing evidence as pairwise comparisons rather than absolute scores is classical in ranking theory. Given comparisons $u \succ v$, two largely separate traditions convert them into a ranking. The first is *probabilistic preference modeling*: Bradley–Terry-type models explain observed comparison outcomes through latent per-item strengths estimated by maximum likelihood [5, 22], and Plackett–Luce models generalize the same idea to full or partial orderings [32, 40]. Rank Centrality recovers a ranking as the stationary distribution of a Markov chain built from pairwise comparisons, without fitting a parametric strength model [36], and combinatorial Hodge theory decomposes a pairwise comparison graph into a gradient (globally consistent) component and a curl/harmonic (locally or globally cyclic) residual, giving a spectral account of how far the data are from admitting a single ranking [25, 52]. Bias-aware pairwise ranking further shows that even the sampling process that generates which pairs are compared can itself introduce systematic ranking bias, independent of any repair step [16].

The second tradition is *combinatorial rank aggregation and cycle repair*, which treats a set of pairwise preferences (or several ranked lists) as combinatorial input and asks for an ordering, or a modification of the input, that minimizes a specific disagreement objective. Kemeny’s optimal-consensus ranking minimizes total pairwise disagreement with a set of input rankings [27]; web-scale rank aggregation work popularized Markov-chain, Copeland-style (wins-minus-losses), and Borda-style (positional) aggregation rules and analyzed their properties [11]; and follow-up work gave approximation algorithms and complexity results for Kemeny-optimal and related aggregation problems, connecting them to correlation clustering [1, 28]. Rank comparison metrics that tolerate ties and partial overlap, such as the Fagin–Kumar family, are a companion line of work for measuring how much two rankings (or a ranking before and after a repair step) actually differ [14, 15].

These two traditions answer different questions. Probabilistic models estimate a ranking that best explains noisy comparison outcomes and treat residual disagreement

as statistical noise around a latent order. Combinatorial repair instead treats the observed graph as fixed evidence and asks how to modify it, at minimum cost, so that a single consistent order becomes possible. The present paper is positioned entirely in the second tradition: it uses combinatorial minimum-weight feedback-arc-set repair (Section 3.4) on an already-constructed preference graph, and its research question is not about estimating latent item strengths but about whether the combinatorial repair objective, once optimized, is diagnostic of a different objective – retrieval effectiveness – that it was never designed to optimize.

2.2 Pairwise and Graph-Based Ranking in Information Retrieval

Pairwise comparison has become a common building block of modern retrieval and reranking pipelines. Pairwise ranking prompting elicits document-level comparisons directly from a large language model and aggregates them into a ranking [42], and pointwise, pairwise, and listwise LLM reranking paradigms have been compared directly for effectiveness and cost [48]. Listwise and setwise rerankers reduce the number of language-model calls needed to reorder a candidate list [18, 41, 58, 62]. Hybrid rerankers combine lexical, dense, and cross-encoder signals within a single scoring or fusion step [31, 37, 54, 60]. Graph-free score and rank fusion – reciprocal rank fusion, CombSUM, and Borda-style combination – remains a strong and widely used alternative to any graph-based aggregation step [9, 17, 30, 53]. Such methods also avoid comparing incompatible raw score magnitudes, a long-standing concern in fusion and score-normalization work [3, 6, 35].

A smaller body of work builds an explicit preference graph from pairwise document judgments before extracting a ranking, rather than fusing scores or ranks directly. **PRP-Graph** builds a ranking graph from pairwise LLM prompting and aggregates it for text reranking [33]; **LLM-RankFusion** aggregates several LLM rankers by explicitly modeling and mitigating pairwise inconsistency in the resulting comparisons [59], and confidence-aware or bias-aware pairwise aggregation methods weight or correct individual comparisons before they are combined [16]. This is the setting the present paper studies: a query-specific weighted directed graph built from several ranking signals, from which a ranking is extracted before or after a structural repair step. Standard retrieval benchmarking practice – reusing stored candidate pools and score files rather than re-running upstream retrieval, and evaluating with rank-sensitive metrics such as nDCG – follows the same evaluation conventions established for zero-shot retrieval benchmarking more generally [10, 49].

2.3 Inconsistency, Cycles, and Preference Repair

A directed preference graph admits at least one total order consistent with all retained edges if and only if it is acyclic. Acyclicity does not imply that the observed edge relation is transitively closed: a DAG need not equal its transitive closure, and several linear extensions may remain compatible with the same retained edges. In general aggregated preference graphs are not acyclic: combining partially disagreeing sources routinely produces directed cycles, including simple two-way contradictions

and longer cyclic chains. The combinatorial problem of removing the least-weight set of edges needed to make a weighted directed graph acyclic is the (minimum-weight) feedback arc set problem. It is NP-hard in general [26]; classical work gives fast, deterministic greedy heuristics that repeatedly remove the lightest edge on a discovered cycle [12], and later work gives polynomial-time approximation schemes and connects the problem to correlation clustering [1, 28]. Exact approaches are also commonly formulated through the related linear-ordering problem and branch-and-cut/cutting-plane methods [2, 19]. This paper does not claim the minimum-weight feedback-arc-set formulation itself as a contribution – it is a long-studied classical combinatorial optimization problem – and instead studies the empirical consequence of solving it, both heuristically and to certified optimality, for a downstream retrieval task.

The same structural issue has recently reappeared in the context of large language models used as pairwise or comparative judges: repeated or multi-evaluator LLM comparisons frequently produce transitivity violations and directed cycles, which several recent studies diagnose or attempt to resolve [21, 46, 61]. That work generally treats acyclicity (or reduced contradiction) as a desirable property of the resulting evaluation and reports downstream gains on tasks such as model ranking, response selection, or fine-tuning-data selection [21]; it does not test whether the acyclicity objective itself is diagnostic of retrieval effectiveness under a corrected, paired statistical protocol, which is the specific gap this paper addresses.

This paper substantially extends and revises an earlier preprint by the same author [51], which studied a smaller and partially overlapping evidence base under a less conservative evaluation protocol (no separation of canonical from larger-pool evaluation cells, no Holm correction applied to the headline retrieval claim, and no exact-repair comparison) and reported a conditional, dataset-dependent positive retrieval effect. That preprint has not completed peer review at its original target venue and is cited here only as a preprint, not as a prior peer-reviewed publication. The present study is not a resubmission of it: it uses a materially more conservative statistical protocol, adds exact SCIP-based repair specifically to rule out heuristic-repair artifacts as an explanation, and reaches a different, more restrained conclusion on an updated evidence base. Readers should treat the two as distinct, non-comparable analyses rather than as an earlier and later version of the same result.

2.4 Structural Quality Versus Downstream Utility

A recurring pattern across the work reviewed above is that structural or intrinsic quality – cyclicity, removed feedback-arc weight, agreement with a probabilistic comparison model, or acyclicity of an aggregated evaluation graph – is measured and reported, but rarely tested directly against an independent downstream task metric under corrected statistical inference. This mirrors a broader distinction in the data-quality literature between *intrinsic* quality (properties of the data or artifact itself) and *contextual* or task-relative quality (fitness for a specific use) [55]; work on data cascades and benchmark label error shows concretely that upstream curation or construction choices can shape downstream conclusions in ways that are easy to miss when only the final task metric is inspected [38, 45]. Simple significance testing is also known to be an unreliable guide to retrieval system comparisons unless multiplicity

and pairing are handled correctly [7, 23, 50], and the interpretation of a non-significant result requires care distinct from ordinary significance testing [29].

Few studies in the preference-graph or feedback-arc-set literature isolate structural improvement from downstream task utility as two separately measured and separately tested quantities. Most either report only structural diagnostics, or report a task metric without a repair-method control (e.g. exact repair) that would rule out heuristic suboptimality as the explanation for a null or weak downstream effect. The controlled separation used here – explicit construction, unrepaired versus heuristic versus exact repair, and multiplicity-corrected paired testing of the structural-to-retrieval link – is designed to close that specific gap (Section 1).

3 Background and Problem Formulation

This section fixes notation and formalizes the four stages distinguished in Section 1 – retrieval setting, derived preference graph, structural inconsistency, and repair – before turning to ranking extraction and evaluation. Every definition below is stated at the level of generality needed to state the paper’s research question precisely; the specific benchmarks, ranker families, and evaluated cells that instantiate this formulation are given in Section 4.

3.1 Retrieval and Candidate Rankings

Let $q \in \mathcal{Q}$ denote a query drawn from a fixed query set, and let $D_q = \{d_1, \dots, d_n\}$ denote its query-specific candidate document set: the pool of documents that will be compared and ranked for q . Candidate sets are formed upstream of this study’s pipeline, typically as the union of documents retrieved or scored by one or more base rankers, and are held fixed once formed.

A base ranker (or, more generally, a judgment source) $s \in S$ assigns a score $x_{q,s}(d) \in \mathbb{R}$ to each document $d \in D_q$ it scores; a ranker need not score every document in D_q , so scores may be missing for some (q, s, d) triples. Because different rankers’ raw scores are not generally comparable in scale or distribution, scores are normalized per query and per ranker before they are compared across rankers. The normalization used throughout the canonical experiments is per-query, per-ranker min-max scaling,

$$\bar{x}_{q,s}(d) = \frac{x_{q,s}(d) - \min_{d' \in D_{q,s}} x_{q,s}(d')}{\max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d')}, \quad (1)$$

where $D_{q,s} \subseteq D_q$ is the subset of candidates actually scored by ranker s on query q . If all retained scores for a query–ranker pair are equal (range below 10^{-12}), the normalized vector is set to all zeros; missing candidates are excluded from the range calculation and remain abstentions when votes are formed. Section 3.2 builds pairwise votes from $\bar{x}_{q,s}$ rather than from $x_{q,s}$ directly; the raw, unnormalized construction is retained only as an explicit robustness ablation, not as an alternative primary protocol, precisely because raw heterogeneous score margins are not a scale-neutral starting point for comparing rankers.

Relevance judgments (qrels) associate each query-document pair with a graded or binary relevance label, drawn from each benchmark’s original human-annotated judgments; they are external to, and never derived from, any ranker score or preference graph. All retrieval metrics reported in this paper are computed against these external judgments, evaluated at a fixed prefix length k (the evaluation cutoff): a metric such as $\text{nDCG}@k$ scores only the top- k documents of a candidate ranking. Because the candidate pool D_q evaluated by a ranking method may be larger than the cutoff k , the relationship between the pool size $P = |D_q|$ (or a truncated evaluation pool size) and k is itself an explicit design choice returned to in Section 4: repair can only change which documents occupy the top- k set when $P > k$ gives it room to do so.

3.2 Derived Weighted Preference Graph

For each query q , the normalized ranker scores $\bar{x}_{q,s}$ are converted into a weighted directed preference graph

$$G_q = (V_q, E_q, w_q), \quad V_q = D_q, \quad (2)$$

in which a directed edge $(u, v) \in E_q$ records retained evidence that document u should be ranked ahead of document v , and the edge weight $w_q(u, v) > 0$ records the strength of that retained evidence. The graph is a *derived* data object, not ground truth: it is the output of an explicit construction procedure applied to the normalized scores, and its structure – which edges exist, in which direction, and with what weight – depends on that procedure’s choices, not on any property of the documents themselves.

Construction proceeds ranker by ranker and pair by pair. For each ranker s and each unordered document pair $\{u, v\} \subseteq D_q$ that s scores, s casts a directional vote for whichever of $u \succ v$ or $v \succ u$ its normalized scores support, with vote strength derived from the normalized score margin $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$. A pair with an exactly equal normalized score, or for which s does not score both documents, abstains rather than casting an imputed vote: ties and missing comparisons are treated as an absence of evidence, not as evidence of indifference. Votes across rankers are then aggregated per directed pair, and a pair-level and graph-level retention rule (the *vote-construction regime*) decides which directions are actually retained as edges of G_q .

This paper studies three explicit vote-construction regimes that hold the candidate set and underlying ranker scores fixed and vary only the retention rule:

- **ms2** (conservative): retain a direction only when it has majority-style support across rankers and the aggregated margin exceeds a fixed threshold. This suppresses most direct two-way contradictions and, in the canonical package, produces graphs that are nearly always acyclic.
- **ms1** (permissive): retain every ranker-supported directional vote. Because different rankers may disagree, both (u, v) and (v, u) can be retained for the same unordered pair, producing direct mutual contradictions and, transitively, longer cycles.
- **ms1_drop_mutual**: like **ms1**, but when both directions of an unordered pair would otherwise be retained, drop the pair entirely at vote construction (before aggregation). This isolates how much of **ms1**’s cyclicity is attributable specifically to direct mutual contradiction rather than to longer-cycle nontransitivity; it is not a post-hoc

rewrite of an already-built `ms1` graph, and retention-matched aggregate thresholds may differ from `ms1` (Section 4.5).

These three regimes are not alternative proposals for a “correct” construction rule; they are a controlled sweep whose purpose is to make graph construction, rather than the repair algorithm, a first-class experimental factor (Section 4 reports how strongly this factor determines observed cyclicity). Because a conflicting vote-construction regime can produce a graph in which repair has little or no structural work to do, an observed repair effect (or its absence) cannot be interpreted without knowing which regime produced the graph it was applied to.

3.3 Cycles and Structural Inconsistency

A weighted directed graph G_q is *acyclic* if it contains no directed cycle $u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_m \rightarrow u_1$; equivalently, its strongly connected components (maximal vertex sets within which every vertex is reachable from every other) are all singletons. G_q admits at least one linear extension consistent with every retained edge direction if and only if it is acyclic; a cyclic G_q cannot be represented exactly by any single total order over D_q ; some retained edge must be violated by any linear ranking of the graph’s vertices.

This paper reports two purpose-specific structural inconsistency measures, rather than a single omnibus cyclicity score. The first is a per-query binary indicator of whether G_q is cyclic at all under a given construction regime, aggregated across queries into a cyclic-query rate. The second decomposes that rate into cyclicity attributable to direct mutual contradiction (a pair retained in both directions) versus cyclicity that survives after mutual pairs are removed. Two related diagnostics are reported: a post-aggregation deletion of reciprocal edges on an `ms1` graph, and the separate `ms1_drop_mutual` construction that drops contested pairs at vote time. These are not guaranteed to coincide (Section 5.1), because they apply mutual handling at different stages and may use different retention-matched aggregate thresholds; conflating them would over- or under-state how much of the observed cyclicity a repair step is actually resolving.

Acyclicity is a purely structural property: it guarantees that G_q admits *some* consistent total order, but it does not by itself guarantee that the retained edges are correct, that the graph’s implied ranking agrees with relevance judgments, or that any particular extraction rule applied to an acyclic graph will outperform one applied to a cyclic graph. Transitivity, acyclicity, evidential correctness, and retrieval optimality are four distinct properties; this paper treats none of them as substitutes for any other, and Sections 3.4 and 3.5 keep the objective that repair optimizes (feedback-arc weight) explicitly separate from the metric evaluation uses (nDCG against qrels).

3.4 Minimum-Weight Feedback-Arc-Set Repair

Given a (possibly cyclic) preference graph $G_q = (V_q, E_q, w_q)$, repair seeks a minimum-weight set of edges $F_q^* \subseteq E_q$ whose removal leaves an acyclic graph:

$$F_q^* = \arg \min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{s.t.} \quad (V_q, E_q \setminus F) \text{ is acyclic.} \quad (3)$$

This is the minimum-weight feedback arc set (MWFAS) problem (Section 2.3); the repaired graph is $\tilde{G}_q = (V_q, E_q \setminus F_q^*, w_q)$. The objective in Equation (3) is entirely graph-internal: it minimizes disagreement with G_q 's own retained edges, subject to the feasibility condition that the result be acyclic. It contains no reference to relevance judgments, ranking metrics, or any notion of retrieval utility – a property that is central to this paper's research question rather than an incidental modeling detail.

Two solution methods for Equation (3) are used and reported as co-equal, not as a primary method with a secondary check. **Greedy repair** is a fast, deterministic heuristic: while $\tilde{G}_q$ contains a directed cycle, find one such cycle and remove its minimum-weight edge, repeating until the graph is acyclic. This does not generally reach the optimum of Equation (3), but it is cheap enough to run at the scale of the full query set. **Exact repair** solves Equation (3) to certified optimality via a standard linear-ordering mixed-integer program [2, 19]: the MWFAS problem can equivalently be written as a linear-ordering problem – choose a total order and minimize the weight of edges directed backward with respect to that order. For every ordered pair of nodes (u, v) , a binary variable $\text{before}[u, v]$ indicates that u precedes v in the chosen total order, subject to antisymmetry ($\text{before}[u, v] + \text{before}[v, u] = 1$) and transitivity constraints over every ordered triple of nodes; the objective is

$$\min \sum_{(u,v) \in E_q} w_q(u, v) \text{before}[v, u], \quad (4)$$

i.e. the total weight of retained edges oriented against the order. The removed set is exactly those backward edges, so the objective equals the MWFAS objective in Equation (3); every acyclic retained subgraph admits at least one topological ordering, and the MIP searches over all total orders. This program is solved with the open-source SCIP solver; a query graph is included in the exact-repair evidence only when the solver certifies proven optimality, not merely a feasible incumbent.

Exact repair is used here as a *diagnostic control*, not as a proposed practical replacement for greedy repair: it does not scale to the largest query graphs, and this paper makes no claim about the general scalability or practicality of exact MWFAS solving. Its role is narrower and specific to this study's question. Because F_q^* from exact repair is, by construction, a global optimum of Equation (3), no repair method can remove less weight while still restoring acyclicity. If exact repair's downstream retrieval effect is still statistically indistinguishable from greedy repair's, then the suboptimality of the greedy heuristic is ruled out as an explanation for a null or weak retrieval effect – precisely the objection a greedy-only study cannot itself close (Section 1).

3.5 Ranking Extraction and Evaluation

A final query-specific ranking is extracted from either the unrepaired graph G_q or the repaired graph $\tilde{G}_q$, or from the upstream ranker signal alone. The canonical experiments compare four families of extraction rule, evaluated in both unrepaired and repaired form where the rule is graph-dependent:

- **Prior-only:** rank by the upstream per-document score (e.g. a normalized score sum across rankers), independent of any graph.
- **Graph-free fusion:** reciprocal rank fusion, CombSUM, and Borda-style fusion over the constituent rankers’ individual rankings, which likewise does not construct or consult a preference graph.
- **Graph-only extraction:** a Copeland-style score $s_{\text{cop}}(u) = \deg^+(u) - \deg^-(u)$ (unweighted out-degree minus in-degree); a weighted balance score $s_{\text{bal}}(u) = \sum_{(u,v) \in E} w(u,v) - \sum_{(v,u) \in E} w(v,u)$ (weighted out-edge mass minus in-edge mass); and one or more stationary-distribution scores computed from a Markov chain over the (weighted) graph, in the spirit of Rank Centrality (Section 2.1). Section 4 gives the precise, distinct instantiations this study evaluates (a damped chain and an undamped Rank Centrality chain are both used, as two related but separately reported methods, not interchangeable names for one method) together with the further graph-centrality and probabilistic-comparison extraction rules (PageRank, Bradley–Terry, topological orderings) included in specific comparisons. Each graph-only rule is computed on both G_q (unrepaired) and $\tilde{G}_q$ (repaired) wherever both are defined, holding the extraction rule fixed so that any difference is attributable to repair rather than to a change of scoring rule.
- **Hybrid prior-plus-graph extraction:** the graph score is combined with a graph-free prior score,

$$h_q(d) = \hat{s}_q^{\text{prior}}(d) + \alpha \hat{s}_q^{\text{graph}}(d), \quad (5)$$

where both components are independently min–max normalized query-wise (constant components map to zeros) and the primary analysis fixes $\alpha = 0.3$ (with a sweep $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ reported as a robustness check). The prior $\hat{s}_q^{\text{prior}}$ is reciprocal rank fusion over the three base rankers on D_q with $k=60$, not a simple score sum; final ties break by document id. When $\hat{s}_q^{\text{graph}}$ is computed from G_q the method is unrepaired; when computed from $\tilde{G}_q$ it is repaired.

Retrieval effectiveness is measured with normalized discounted cumulative gain (nDCG), the standard graded-relevance ranking metric [24]: for a ranking with relevance labels $\text{rel}_1, \dots, \text{rel}_k$ at ranks $1, \dots, k$,

$$\text{DCG}@k = \sum_{i=1}^k \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}, \quad \text{nDCG}@k = \frac{\text{DCG}@k}{\text{IDCG}@k}, \quad (6)$$

where $\text{IDCG}@k$ is the DCG of the ideal (qrels-sorted) ranking at the same cutoff, so $\text{nDCG}@k \in [0, 1]$ with higher values indicating better agreement with the relevance judgments. nDCG is computed per query and compared between the unrepaired and repaired variant of the same extraction rule, holding every other choice fixed; Section 4 specifies the paired, multiplicity-corrected inference protocol used to test whether an observed repaired-versus-unrepaired difference is reliable.

The formulation above makes explicit why improving Equation (3) is not, by construction, the same as improving nDCG. The repair objective is computed entirely from G_q ’s own edges and weights, which are themselves a derived function of upstream

normalization, vote semantics, and thresholding choices (Section 3.2); nDCG is computed from an extraction rule’s output ranking against relevance judgments that never enter the repair objective at all. A smaller feedback-arc weight, or proven-optimal removal of that weight, is evidence about internal consistency of the retained votes – it is not, on its own, evidence about agreement with relevance judgments. Whether the two are empirically linked in this study’s evidence, and how reliably, is the question Sections 4 and 5 take up directly.

4 Methodology

Section 3 formalized the objects this study reasons about – candidate rankings, the derived preference graph, structural inconsistency, MWFAS repair, and ranking extraction – at the level of generality needed to state the paper’s research question precisely. This section instantiates that formulation: the specific research questions it answers, the concrete pipeline, datasets, rankers, construction protocol, repair implementations, extraction methods, baselines, metrics, and statistical-inference procedure used to produce the evidence reported in Section 5. No outcome values are reported here; this section documents what was measured and how, not what was found.

4.1 Research Questions

The study answers four questions that operationalize the separation of construction, structural repair, and retrieval utility (Section 1):

RQ1 How much structural inconsistency – cyclicity and contradictory edge mass – is present in derived multi-ranker preference graphs, and how strongly does it depend on graph-construction choices (Section 3.2) rather than on the underlying documents?

RQ2 How do greedy and exact minimum-weight feedback-arc-set repair (Section 3.4) differ in the structural change they produce – removed edge weight, the chosen feedback arc set, and top- k membership when the candidate pool exceeds the evaluation cutoff – on the same graphs?

RQ3 Does preference-graph repair, heuristic or exact, produce a statistically reliable improvement in downstream retrieval effectiveness (nDCG) relative to the unrepaired graph, once paired inference and multiplicity correction are applied – and how do repaired graph methods compare with simple graph-free fusion anchors?

RQ4 Is any observed retrieval null in RQ3 attributable to greedy-repair suboptimality rather than to repair itself, once exact MWFAS is used as a diagnostic control under matched construction, pooling, and extraction choices?

These questions concern the marginal effect of repair under the stated protocol. They do not ask for a learned per-query policy that decides when to invoke repair, nor for a state-of-the-art reranker comparison beyond the graph-free fusion anchors used to contextualize that marginal effect.

4.2 Experimental Pipeline

Every reported result is produced by the same nine-stage pipeline, run once per query under each combination of dataset, vote-construction regime, and evaluation cell studied:

1. obtain stored per-document scores from each base ranker (Section 4.4) for a fixed candidate pool;
2. normalize scores per query and per ranker (Equation (1));
3. derive weighted pairwise votes from normalized score margins (Section 3.2);
4. construct the query-specific weighted directed preference graph G_q under one vote-construction regime (Section 4.5);
5. extract unrepaired rankings directly from G_q and from the prior score alone;
6. repair G_q into $\tilde{G}_q$ with greedy cycle-peeling or exact SCIP-based MWFAS solving (Section 4.6);
7. extract repaired rankings from $\tilde{G}_q$ using the same extraction rule applied to G_q (Section 4.7);
8. evaluate every extracted ranking against relevance judgments with nDCG at a prespecified cutoff (Section 4.9); and
9. conduct paired, multiplicity-corrected statistical inference on repaired-versus-unrepaired differences (Section 4.10).

Stages 1–4 are shared by every downstream comparison; stages 5–9 are repeated once per extraction rule and once per repair method (unrepaired, greedy, exact) being compared. No stage regenerates upstream retrieval: stage 1 reuses fixed, versioned score files, so every reported statistic is reproducible from stored intermediates without repeating retrieval or querying a paid API (Section 4.11).

Figure 1 summarizes this pipeline, with the greedy and exact repair branches drawn explicitly as parallel paths through the same downstream extraction and evaluation stages, so that any difference between repaired and unrepaired rankings, or between greedy- and exact-repaired rankings, is attributable to the repair step alone.

4.3 Datasets and Retrieval Tasks

The study uses four public retrieval benchmarks, chosen to span distinct retrieval domains so that construction-sensitivity and repair findings can be checked for domain generality rather than being tied to one corpus type: SciDocs, a scientific-document retrieval benchmark built around citation-informed document representations [8]; FiQA, a financial-opinion question-answering and retrieval benchmark [34]; HotpotQA, a multi-hop question-answering benchmark used here as a supporting-fact retrieval task [57]; and BRIGHT, a reasoning-intensive retrieval benchmark whose queries require inference beyond lexical or semantic similarity to identify relevant documents [47], accessed via its public `examples` configuration. SciDocs, FiQA, and HotpotQA also appear in general zero-shot retrieval benchmarking [49]; BRIGHT is included specifically for a more recent, reasoning-oriented retrieval setting not covered by that benchmark suite.

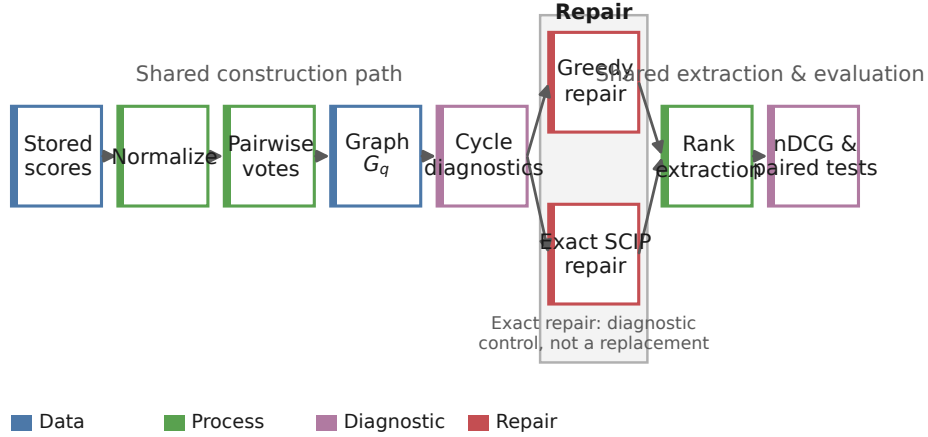


Fig. 1 Pipeline schematic. Shared construction stages produce a preference graph that is diagnosed for cycles and then either left unrepaired or repaired by greedy cycle-peeling or exact SCIP MWFAS solving as parallel branches. Both branches feed the same ranking extraction and evaluation stages so that repaired-versus-unrepaired and greedy-versus-exact comparisons hold every downstream choice fixed.

For every dataset, base-ranker scores, query-ID lists, and relevance judgments are reused unchanged from fixed, versioned stored files; no upstream retrieval is regenerated for this study. Queries are filtered only for relevance-judgment eligibility (a query must have at least one qrels-eligible relevant document among candidates that at least one ranker scores); this excludes 18 of HotpotQA’s 70 stored queries and leaves every other dataset’s stored queries usable. Candidate pools are formed, per query, from the union of documents scored by at least one base ranker, truncated to a fixed depth by the primary candidate-pool policy (Section 4.5); Table 1 reports the verified common stored score depth per dataset and the evaluation cells this study evaluates.

4.4 Base Rankers and Judgment Sources

Three base rankers provide heterogeneous ranking signals for the preference-aggregation study; they are a controlled substrate for studying construction and repair, not a claim of retrieval state of the art. **BM25** scores each candidate with the standard Okapi BM25 term-weighting formula [43], using the conventional parameter values $k_1 = 1.5$, $b = 0.75$. **TF-IDF** scores candidates by cosine similarity between query and document term-frequency-inverse-document-frequency vectors under standard logarithmic term-frequency weighting [44]. **MiniLM** scores candidates by cosine similarity between query and document embeddings from the pretrained **all-MiniLM-L6-v2** sentence-embedding model, a distilled transformer encoder [56] used here as a fixed, off-the-shelf dense-similarity signal, not fine-tuned for any dataset in this study. All three rankers are applied to the same per-query candidate pool and

Table 1 Datasets and prespecified evaluation settings. Stored: usable query count after relevance-judgment filtering. Depth: verified common complete stored score depth across all three base rankers. Canonical cells use pool size P equal to the metric cutoff k ; larger-pool cells use $P > k$, so that repair has room to change top- k set membership (Section 3.4).

Dataset	Usable	Depth	Evaluated (P, k) cells
SciDocs	120	50	canonical $(20, 20)$; larger-pool $(20, 5)$, $(20, 10)$, $(20, 20)$, $(50, 5)$, $(50, 10)$, $(50, 20)$
FiQA	120	50	same as SciDocs
HotpotQA	52	35	canonical $(10, 10)$; larger-pool $(10, 5)$, $(35, 5)$, $(35, 10)$, $(35, 20)$
BRIGHT	50	50	same as SciDocs

produce one real-valued score per candidate; higher scores indicate stronger estimated relevance for every ranker, so no ranker requires a direction transformation before normalization (Equation (1)).

4.5 Preference-Graph Construction

The primary construction protocol normalizes every ranker’s scores with per-query, per-ranker min-max scaling (Equation (1)) and extracts directional votes from normalized score margins, following Section 3.2. Vote retention uses a retention-matched thresholding rule that never consults qrels or nDCG. On each dataset, a raw-margin (unnormalized) construction with the same support rule supplies targets: per-ranker retained-vote rates and the regime’s retained directed edge count. For the normalized protocol, each ranker’s vote-margin threshold is the empirical quantile of that ranker’s pooled pairwise normalized margins matching the raw retained-vote rate ($q = 1 - \text{target_rate}$). Given those vote thresholds, the regime-level aggregate weight threshold is chosen by a discrete search over candidate aggregate weights to minimize the absolute gap in retained edge count versus the raw baseline for that regime (ties break toward the smaller threshold). Vote-margin thresholds are therefore per dataset and per ranker, shared across regimes within a dataset; aggregate thresholds are per dataset and per regime; neither is per query. Thresholds are fixed from the full usable query set of that dataset before retrieval metrics are computed. Matching is approximate (minimum absolute edge-count gap), not exact equality of every rate. Canonical values appear in Table 2. Support requirements are fixed per regime: `ms2` requires support from at least two of the three rankers; `ms1` and `ms1_drop_mutual` require support from at least one ranker, differing in whether contested (two-way) pairs are dropped at vote construction (Section 3.2).

Candidate pools use one primary policy: for each query, the top- P documents of a reciprocal-rank-fusion ranking computed over the union of the three rankers’ native (unnormalized) score lists. This “RRF-centered” pool policy is one specific, named choice among several equally valid ways to truncate a candidate pool, and it determines which documents are eligible to appear in D_q at all – a decision made once,

Table 2 Retention-matched thresholds for the primary protocol (`primary_minmax_retention_matched`). Vote thresholds are shared across regimes within a dataset; aggregate thresholds and drop-mutual flags are regime-specific. Matching target: raw-margin retained-vote rates (per ranker) and raw retained edge count (per regime). Calibration uses the same usable queries later evaluated; qrels/nDCG are not used. Regime abbreviations: `ms1`, `ms1dm=ms1_drop_mutual`, `ms2`.

Dataset	Reg.	BM25	MiniLM	TF-IDF	Agg.	Sup.	Drop
SciDocs	<code>ms1</code>	0.00468	0.274	0.340	0.0108	1	no
SciDocs	<code>ms1dm</code>	0.00468	0.274	0.340	0	1	yes
SciDocs	<code>ms2</code>	0.00468	0.274	0.340	0	2	no
FiQA	<code>ms1</code>	0.00622	0.276	0.375	0.0120	1	no
FiQA	<code>ms1dm</code>	0.00622	0.276	0.375	0	1	yes
FiQA	<code>ms2</code>	0.00622	0.276	0.375	0	2	no
HotpotQA	<code>ms1</code>	0.00317	0.272	0.228	0	1	no
HotpotQA	<code>ms1dm</code>	0.00317	0.272	0.228	0.00341	1	yes
HotpotQA	<code>ms2</code>	0.00317	0.272	0.228	0.295	2	no
BRIGHT	<code>ms1</code>	0.00108	0.240	0.555	0.0114	1	no
BRIGHT	<code>ms1dm</code>	0.00108	0.240	0.555	0.0119	1	yes
BRIGHT	<code>ms2</code>	0.00108	0.240	0.555	0	2	no

upstream of score normalization and graph construction. Four alternative, independently defined pool policies (equal per-ranker depth, round-robin interleaving across rankers, a single-ranker pool, and a CombSUM-ranked pool) are evaluated as a robustness check (Section 5) to test whether the study’s structural and retrieval conclusions depend on this specific choice.

Algorithm 1 summarizes graph construction for a single query, reflecting the procedure implemented in the repository’s graph- and vote-construction code. The description is intentionally compact: the contribution is not a new preference extractor, but a controlled study of how these construction choices affect cyclicity, repair, and downstream retrieval.

4.6 Repair Methods

Two repair methods solve Equation (3), reported as co-equal throughout; their roles are defined in Section 3.4. **Greedy repair** is the repository’s own deterministic cycle-peeling heuristic (not the classical Eades–Lin–Smyth linear-arrangement heuristic [12]; Section 2.3). **Exact repair** solves the linear-ordering mixed-integer program (Equation (4)) with the open-source SCIP solver via PySCIPOpt (version 6.2.1), requiring a proven-optimal status (relative optimality gap 0.0) rather than accepting a feasible incumbent, with a 300-second per-query wall-clock time limit as a safety margin. Every exact-repair result reported in this study reached proven optimality; the time limit was never approached (observed per-query solve times range from sub-millisecond to a few hundred milliseconds at the graph sizes studied here). The linear-ordering formulation was independently cross-checked against an unrelated brute-force exact method (exhaustive permutation search) on synthetic and real query graphs, with all checked cases matching exactly. Exact repair is retained here solely as the diagnostic control defined in Section 3.4, answering RQ4 (Section 5.4).

Algorithm 1 Preference-graph construction for one query

Require: Query q ; candidate set D_q ; base rankers $S = \{\text{BM25}, \text{TF-IDF}, \text{MiniLM}\}$ with scores $x_{q,s}$; construction regime $r \in \{\text{ms2}, \text{ms1}, \text{ms1_drop_mutual}\}$

Ensure: Weighted directed preference graph $G_q = (V_q, E_q, w_q)$

- 1: $V_q \leftarrow D_q$; initialize an empty weighted edge accumulator
- 2: **for** each ranker $s \in S$ **do**
- 3: Normalize $x_{q,s}$ to $\bar{x}_{q,s}$ (Equation (1))
- 4: **for** each unordered candidate pair $\{u, v\} \subseteq D_q$ scored by s **do**
- 5: **if** $|\bar{x}_{q,s}(u) - \bar{x}_{q,s}(v)|$ is below the vote-margin threshold, or s does not score both u and v **then**
- 6: Abstain on $\{u, v\}$ for ranker s
- 7: **else**
- 8: Accumulate directional support for $u \succ v$ or $v \succ u$ (whichever $\bar{x}_{q,s}$ supports), weighted by the normalized margin
- 9: **end if**
- 10: **end for**
- 11: **end for**
- 12: Apply regime r 's support rule and aggregate weight threshold to every accumulated directed pair
- 13: **if** $r = \text{ms1_drop_mutual}$ **then**
- 14: Remove every unordered pair retained in both directions
- 15: **end if**
- 16: $E_q \leftarrow$ every retained directed pair with positive aggregated weight
- 17: **return** $G_q = (V_q, E_q, w_q)$

4.7 Ranking Extraction Methods

Extraction methods fall into four groups, extending Section 3.5's general definitions with the specific methods this study evaluates and stating precisely which methods enter which comparison family (Section 4.8).

Prior-only and graph-free fusion rank candidates without constructing or consulting a preference graph: the prior-only method ranks by the same RRF prior used in Equation (5); reciprocal rank fusion, CombSUM, and Borda-style fusion combine the three rankers' individual per-query rankings or scores directly [9, 17].

Graph-only extraction computes a score from G_q or $\tilde{G}_q$ alone. Table 3 records the implementation parameters used throughout. The Copeland score s_{cop} and balance score s_{bal} (Section 3.5) are evaluated on every dataset, regime, and repair method. Two related but distinct Markov-chain scores are also evaluated: a damped stationary-distribution score with uniform restart ("Markov"), and an undamped Rank Centrality chain with explicit self-loops [36]. Standard graph PageRank on the reversed preference graph [39], Bradley-Terry MM MLE with edge weights as win counts [5, 22], and (on repaired DAGs only) NetworkX topological order plus prior-priority topological order complete the set.

Hybrid prior-plus-graph extraction combines a graph-only score with the RRF prior via Equation (5). Copeland and balance hybrids use $\alpha = 0.3$ in the primary

Table 3 Extraction-method implementation parameters matching the canonical pipeline. Direction: edge $u \rightarrow v$ means u preferred over v . Final ranking sorts by descending score then ascending document id unless noted.

Method	Main parameter(s) / ties	Implementation note
Copeland	$\deg^+ - \deg^-$; id tie-break	Isolates score 0
Balance	weighted out-in; id tie-break	Uses edge weights
Markov (damped)	restart 0.15; tol 10^{-10} ; 10^4 iter	Loser→winner + uniform restart
Rank Centrality	self-loops; tol 10^{-8} ; 200 iter	No teleport; normalize π
PageRank	damping 0.85; tol 10^{-6} ; 100 iter	Reversed graph; weighted
Bradley–Terry	MM; tol 10^{-6} ; 100 iter	Uniform init; isolates 0
Topological	NetworkX <code>topological_sort</code>	DAG-only; construction-order ties
Priority-topo	RRF prior ($k=60$) among sources	Higher prior, then larger id
Hybrids	$\alpha=0.3$; min-max each side	$h = \hat{s}^{\text{prior}} + \alpha \hat{s}^{\text{graph}}$; const. $\rightarrow 0$

analysis, with $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ as a robustness check; a Markov hybrid at the same fixed α enters the expanded family (Section 4.8).

As in Section 3.5, every graph-dependent rule is evaluated on both G_q and $\tilde{G}_q$ wherever both are defined. Evaluating graph-only rules alongside hybrids, and reporting the α -sweep, further addresses the concern that a fixed hybrid mixing weight could mask a genuine repair effect at $\alpha = 0.3$.

4.8 Baselines and Comparisons

Table 4 summarizes every method’s role. Two overlapping but distinct method families are evaluated, and Section 5 reports which family backs which specific comparison rather than pooling them: a *primary family* (four graph-free baselines; Copeland, balance, and Markov as graph-only and hybrid rules) backs the primary canonical and larger-pool greedy-repair-versus-unrepaired comparisons; an *expanded family* adds PageRank, Rank Centrality, a Markov hybrid, and Bradley–Terry specifically to the exact-repair-versus-unrepaired canonical and larger-pool comparisons, so that the exact-repair diagnostic control (Section 4.6) is not limited to the extraction rules the primary family happens to use. Topological and priority-topological ordering are evaluated only in a separate exact-versus-greedy structural comparison, since they require an already acyclic graph and so have no unrepaired counterpart.

A “strong baseline” here means the strongest method actually evaluated in this study, not a method added rhetorically: CombSUM and reciprocal rank fusion are simple, graph-free, and require no repair step at all, so their inclusion tests whether any

Table 4 Method families evaluated in this study. Family: which comparison(s) a method backs (Section 5); primary = greedy-repair canonical/larger-pool comparisons; expanded = exact-repair canonical/larger-pool comparisons; struct. = exact-vs-greedy structural comparison only. Repair: which repair variants are compared for that method (U = unrepaired, G = greedy, E = exact).

Method	Family	Graph?	Prior?	Repair variants compared
Prior-only	primary, expanded	no	yes	n/a (no graph)
RRF	primary, expanded	no	no	n/a (no graph)
CombsUM	primary, expanded	no	no	n/a (no graph)
Borda fusion	primary, expanded	no	no	n/a (no graph)
Copeland	primary	yes	no	U, G, E
Balance	primary	yes	no	U, G, E
Markov (damped)	primary	yes	no	U, G, E
Copeland hybrid	primary	yes	yes ($\alpha=0.3$)	U, G, E
Balance hybrid	primary	yes	yes ($\alpha=0.3$)	U, G, E
PageRank	expanded	yes	no	U, E
Rank Centrality	expanded	yes	no	U, E
Markov hybrid	expanded	yes	yes ($\alpha=0.3$)	U, E
Bradley-Terry	expanded	yes	no	U, E
Topological	struct. only	yes	no	G vs. E only
Priority-topological	struct. only	yes	yes (tie-break)	G vs. E only

graph-based method (repaired or not) earns its added complexity relative to methods that discard graph structure entirely. None of the methods in Table 4 is presented as state-of-the-art retrieval; they are a controlled comparison set for isolating the effect of graph construction and repair, not a search for the best possible ranker.

4.9 Evaluation Metrics

The primary retrieval metric is normalized discounted cumulative gain (nDCG) at a prespecified cutoff k (Equation (6)), evaluated per query and averaged across a dataset’s usable queries; higher values indicate better agreement with graded relevance judgments. Mean reciprocal rank (MRR) and mean average precision (MAP) are computed as secondary diagnostics only and are never used as a primary confirmatory metric. All three are computed at the same evaluated (P, k) cell (Table 1) for every method and repair variant being compared.

Following Section 3.5, the repair objective (Equation (3)), structural diagnostics (cyclic-query rate, removed-weight fraction, top- k membership-change rate), and retrieval metrics (nDCG, with MRR/MAP secondary) are never treated as proxies for one another or averaged together.

Table 5 Holm multiplicity families used in this manuscript. Confirmatory families support the primary null claims; robustness families are reported alongside and never substituted for confirmatory Holm decisions. Cell counts match Tables 6 and 8.

Family	Question	Scope	Methods	n
Active canonical	Greedy repair gain?	$4 \times \mathbf{ms1}$, $P=k$	Prim. 5 pairs	20
Full canonical	All regimes?	4×3 regimes, $P=k$	Prim. 5 pairs	60
Active larger-pool	When $P > k$?	$\mathbf{ms1}$, larger-pool	Prim. 5 pairs	110
Exact-repair canonical	Exact vs unrepaired?	$4 \times \mathbf{ms1}$, $P=k$	Exp. 9 pairs	36
Exact-repair larger-pool	Exact vs unrepaired, $P > k$?	under larger-pool cells	Exp. 9 pairs	56
Exact vs greedy	Exact vs greedy ret.?	pooled / per-cell	Prim.+topo	35/399
Alternative-pool	Pool-policy fact?	arti- alt. pools	vs RRF/CombSUM	8
Equivalence (TOST)	Narrow equivalence?	equiva- active larger-pool	Prim. pairs	110

4.10 Statistical Analysis

The unit of statistical inference is the query: for a given dataset, construction regime, evaluation cell, and extraction method, the repaired-minus-unrepaired nDCG difference is computed once per query, and all inference is conducted on this vector of paired per-query differences, never on a coarser or finer unit. The primary confirmatory test for every repaired-versus-unrepaired comparison is an exact sign-flip permutation test on the paired differences (enumerating all 2^m sign patterns when the number of nonzero differences $m \leq 20$; a 10,000-replicate Monte Carlo approximation with a fixed seed otherwise), testing the null hypothesis that the paired differences are symmetric about zero. Every p -value from a reported family of cells (e.g. the full canonical family, or the full larger-pool family) is adjusted jointly with the Holm step-down procedure [20], controlling the family-wise error rate at $\alpha = 0.05$. Families are corrected separately (Table 5): confirmatory families answer the primary repaired-versus-unrepaired nDCG question under greedy or exact repair; robustness families vary pool, baselines, or ask equivalence/power questions. The reported families were fixed in the canonical evidence specification before the final manuscript analysis; family membership was not reopened after inspecting the final Holm decisions. Datasets and regimes within a family remain separate cells (not collapsed into one omnibus test), so repeated regimes from the same query contribute separate paired vectors rather than a hierarchical model. There is no omnibus multi-regime test beyond the separately corrected families listed in Table 5.

Beyond this primary test, four robustness procedures are reported for every primary family, never as a substitute for the Holm-corrected decision: percentile, basic, and bias-corrected-and-accelerated (BCa) bootstrap 95% confidence intervals on the mean paired difference [13]; leave-one-out influence analysis, removing one query at a time to check whether any single query drives a family’s conclusion; the

Benjamini–Hochberg false-discovery-rate procedure [4] as an alternative, less conservative correction, reported alongside (not in place of) the primary Holm decision; and a minimum-detectable-effect (MDE) analysis at 80% power, and two-one-sided-tests (TOST) narrow-equivalence testing at prespecified margins, both used only to characterize what the study’s sample size can and cannot rule out, not to claim equivalence has been established (Section 2.4).

For the appendix-scoped real-large-language-model pilot (Section 1), the unit of inference is the query *cluster*: six queries, each replicated across several provider, paradigm, and repetition conditions, are treated as six independent clusters, never as row-level independent observations. Row-level analysis of this pilot’s replicated rows (correct unit: $n = 6$; incorrect unit: one row per replicate) systematically overstates the pilot’s evidential weight and is not used anywhere in this study.

4.11 Reproducibility and Implementation

The full pipeline (Section 4.2) is implemented in Python, using `networkx` for graph representation and cycle detection, `numpy/scipy` for numerical and statistical routines, and `PySCIP0pt` 6.2.1 (open-source SCIP) for exact repair (Section 4.6); the MiniLM ranker (Section 4.4) uses the `sentence-transformers` library to load the fixed, pretrained `all-MiniLM-L6-v2` checkpoint. Every Monte Carlo or bootstrap procedure (Section 4.10) uses a fixed random seed, so that reported p -values and intervals are exactly reproducible from the same stored inputs. No stage of the pipeline used to produce a main-paper statistic requires a paid API call: base rankers use fixed, versioned, locally computed or stored score files, and the appendix-scoped real-LLM pilot (which does call paid provider APIs) is never a source of a main-paper number. All code, stored intermediates, and the scripts used to regenerate every reported statistic and table are available at the repository referenced in Section 9.

Generative-AI assistance. Generative artificial-intelligence tools (Anthropic’s Claude, via the Claude Code command-line environment) were used to assist with software development for this study – writing and revising analysis code, statistical-inference utilities, figure-generation scripts, and audit scripts – and with drafting, revising, and fact-checking portions of this manuscript’s text and figures. Every AI-assisted code change, statistical analysis, figure, table, citation, and passage of manuscript text was reviewed and independently verified against the underlying stored data and source code by the author before inclusion; no generative-AI output was accepted without that verification. The author takes full responsibility for the accuracy, integrity, and originality of the source code, experiments, statistical analyses, figures, tables, interpretations, citations, and manuscript text in this work. No generative-AI tool is credited as an author, and no generative-AI tool independently designed the study, selected the research questions, or made scientific judgments attributed to the author in this manuscript.

5 Results

This section reports what the canonical evidence package shows for each research question in turn (Section 4.1): how much structural inconsistency the derived preference

graphs contain (RQ1), what repair changes structurally, including top- k membership when $P > k$ (RQ2), whether that change is accompanied by a statistically supported retrieval-effectiveness gain relative to unrepaired graphs and graph-free fusion anchors (RQ3), and whether any retrieval null in RQ3 is attributable to greedy-repair sub-optimality once exact MWFAS is used as a diagnostic control (RQ4). Every statistic below is computed on the canonical `primary_minmax_retention_matched` protocol (Section 4.5) unless stated otherwise, and every significance statement uses the Holm-corrected decision at $\alpha = 0.05$ from the prespecified family it belongs to (Section 4.10); raw, uncorrected p -values are reported only alongside the corrected decision, never as a substitute for it.

5.1 Structural Inconsistency Before Repair

The vote-construction regime, not the underlying documents, determines whether a query’s preference graph is cyclic at all. Figure 2 reports, for every dataset and regime, the fraction of queries whose graph contains a directed cycle, that same fraction after direct mutual contradictions are removed, and the mean feedback-arc weight repair would need to remove, expressed as a fraction of total graph weight, together with the active-regime before/after comparison.

The conservative regime, `ms2`, is acyclic on every query in every dataset in the observed evidence (Section 3.2): repair has no graph to act on under this regime. The permissive regime, `ms1`, is cyclic on the large majority of queries in every dataset, ranging from 63.5% (HotpotQA) to 99.2% (SciDocs). Most of that cyclicity is direct, two-way contradiction rather than longer-range nontransitivity: the construction-time mutual-drop regime (`ms1_drop_mutual`) reduces the cyclic-query rate to between 1.9% (HotpotQA) and 30.8% (FiQA). That rate is close to, but not identical to, the post-hoc “after mutual” diagnostic on `ms1` graphs: the diagnostic deletes reciprocal edges from the already-aggregated `ms1` digraph, whereas `ms1_drop_mutual` drops contested pairs during vote construction and uses its own retention-matched aggregate threshold (Table 2). The two coincide on FiQA, HotpotQA, and BRIGHT, but differ on SciDocs (10.8% post-mutual `ms1` versus 11.7% cyclic under `ms1_drop_mutual`; one query remains cyclic only under the construction-time drop). They therefore measure related but distinct residual longer-cycle structure (Figure 2).

Raw, unnormalized score margins are not a scale-neutral starting point for this construction (Section 3.1). Under the raw-margin ablation, BM25 contributes a mean conditional edge-weight share of 0.988 whenever it participates in an edge; under the primary normalized protocol, that share falls to 0.512. Figure 3 shows this raw-versus-normalized contrast holds across all four datasets and all three vote-construction regimes: a graph built from raw margins is close to a single-ranker graph with multi-ranker labels attached, and normalization is what restores a genuinely mixed graph.

5.2 Structural Effect of Repair

Repair is structurally active precisely where Section 5.1 shows a graph has contradictory structure to remove, and inert elsewhere. Under `ms2`, repair removes nothing

Dataset	Regime	Cyclic (%)	After mutual (%)	FAS wt.
SciDocs	ms2	0.0	0.0	0.000
	ms1	99.2	10.8	0.074
	ms1_drop_mutual	11.7	11.7	0.002
FiQA	ms2	0.0	0.0	0.000
	ms1	98.3	30.8	0.080
	ms1_drop_mutual	30.8	30.8	0.004
HotpotQA	ms2	0.0	0.0	0.000
	ms1	63.5	1.9	0.029
	ms1_drop_mutual	1.9	1.9	0.000
BRIGHT	ms2	0.0	0.0	0.000
	ms1	92.0	22.0	0.055
	ms1_drop_mutual	22.0	22.0	0.003

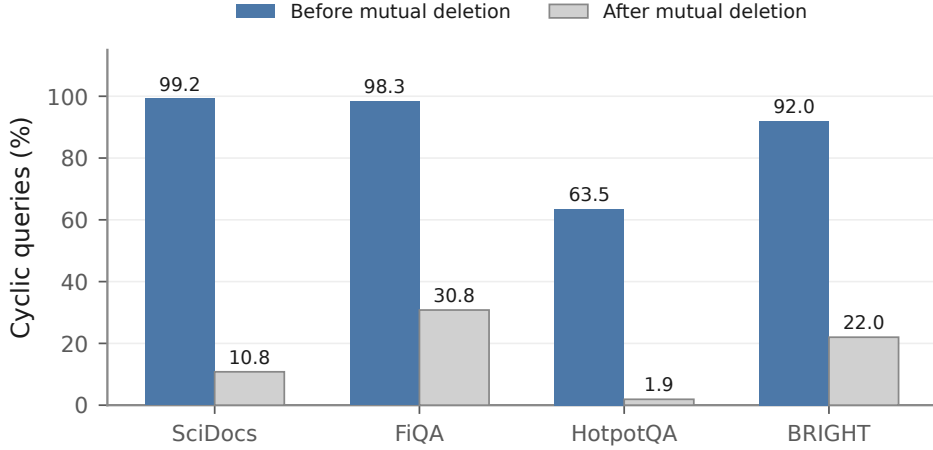


Fig. 2 Structural inconsistency under the canonical protocol. Top: cyclic-query rate, post-mutual diagnostic cyclic rate, and mean normalized feedback-arc weight by dataset and vote-construction regime (After mutual: cyclic rate after deleting reciprocal edges from that regime’s already-aggregated graph; FAS wt.: mean feedback-arc weight removed by repair as a fraction of total graph weight). Bottom: active-regime (ms1) cyclic-query percentage before and after that post-aggregation mutual-pair deletion. Most active-regime cyclicity is direct two-way contradiction, with a nonzero residual after mutual-pair deletion.

on any query, because no graph is cyclic. Under ms1, the active regime, repair removes a mean of 7.4% (SciDocs) to 8.0% (FiQA) of total graph weight; under ms1_drop_mutual, the fraction still to remove after mutual pairs are already gone drops to 0.2%–0.4% (Figure 2). This confirms that graph construction, not the repair algorithm, is what decides whether repair has any opportunity to act at all – a point kept separate here from whether that activity translates into a retrieval effect, which Section 5.3 addresses directly.

Repair also changes which documents are visible for evaluation once the candidate pool exceeds the metric cutoff. At $P = k$, top- k set membership cannot change by construction: the evaluated prefix is the entire pool, so repairing the graph can

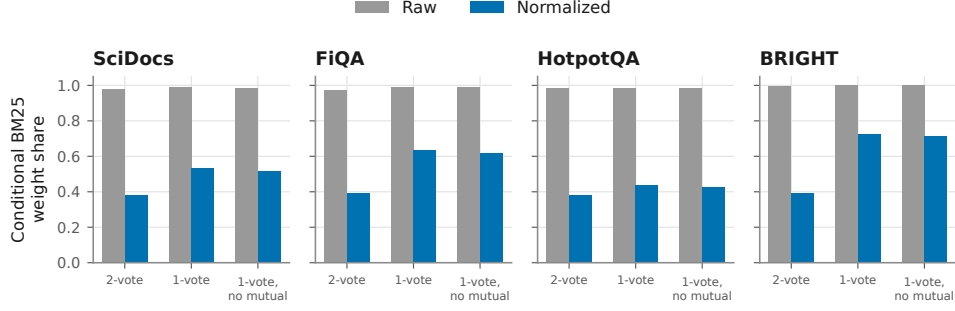


Fig. 3 Conditional BM25 edge-weight share under the raw-margin ablation versus the primary normalized protocol, by dataset and vote regime. Raw construction is near-total BM25 domination in every panel; normalization restores a mixed graph.

reorder documents but never add or remove one from the top- k set, and the measured membership-change rate is exactly 0%. Once $P > k$, pooling across all three vote-construction regimes and every larger-pool evaluation cell (Section 4.3), the mean top- k membership-change rate rises to 10.6%. Repair is therefore not merely reordering an already-fixed prefix under $P > k$: it can and does change which documents are evaluated at all. Whether that structural activity is accompanied by a reliable retrieval-effectiveness gain is addressed next.

5.3 Retrieval Effectiveness

Table 6 reports the Holm-corrected outcome for every prespecified repaired-versus-unrepaired nDCG comparison family under greedy repair, together with dataset-macro mean nDCG for the strongest graph-free baselines and the best-performing repaired hybrid, for context.

No repaired-versus-unrepaired nDCG comparison survives Holm correction in any of the three families in Table 6 (smallest Holm-adjusted $p = 0.240$ in the active canonical family; SciDocs Copeland hybrid, raw sign-flip $p = 0.012$ before correction). Figure 4 shows the corresponding paired bootstrap intervals for active ms1: point estimates are sometimes positive, but nearly every interval crosses zero.

Simple graph-free baselines remain competitive with the best repaired graph-dependent method throughout: CombSUM’s dataset-macro mean nDCG (0.554) exceeds every graph-dependent method’s, and reciprocal rank fusion and the repaired Copeland hybrid have the same reported dataset-macro mean to three decimal places (0.546). Repair is therefore not shown to produce a retrieval ranking that outperforms simple score or rank fusion under the canonical protocol.

5.4 Exact Versus Heuristic Repair

The comparisons in Section 5.3 use greedy repair, which does not generally reach the optimum of the feedback-arc-set objective (Equation (3)). This subsection reports

Table 6 Retrieval effectiveness under greedy repair, canonical protocol. Top: Holm-corrected repaired-vs-unrepaired nDCG comparison families (5-pair primary method family, Section 4.8). Bottom: dataset-macro mean nDCG for selected methods, for context (not a significance claim; graph-free baselines require no repair step at all).

Comparison family	Cells	Holm-rejected	Smallest Holm-adjusted p
Active canonical (ms1)	20	0/20	0.240
Full canonical (all regimes)	60	0/60	0.720
Active larger-pool (ms1 , $P > k$ incl.)	110	0/110	0.352

Method (dataset-macro mean nDCG)	Mean nDCG
CombSUM (graph-free fusion)	0.554
Reciprocal rank fusion (graph-free)	0.546
Copeland hybrid, repaired (ms1)	0.546
Prior-only (graph-free)	0.545
Borda fusion (graph-free)	0.545

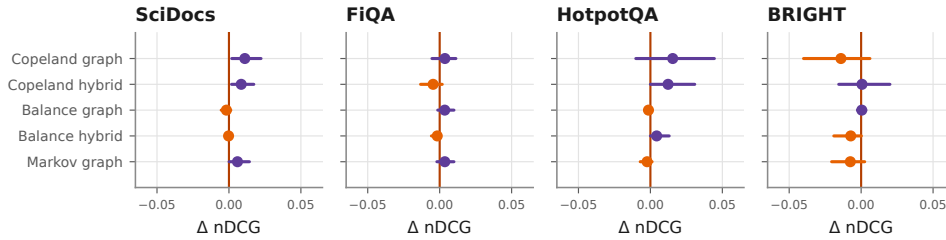


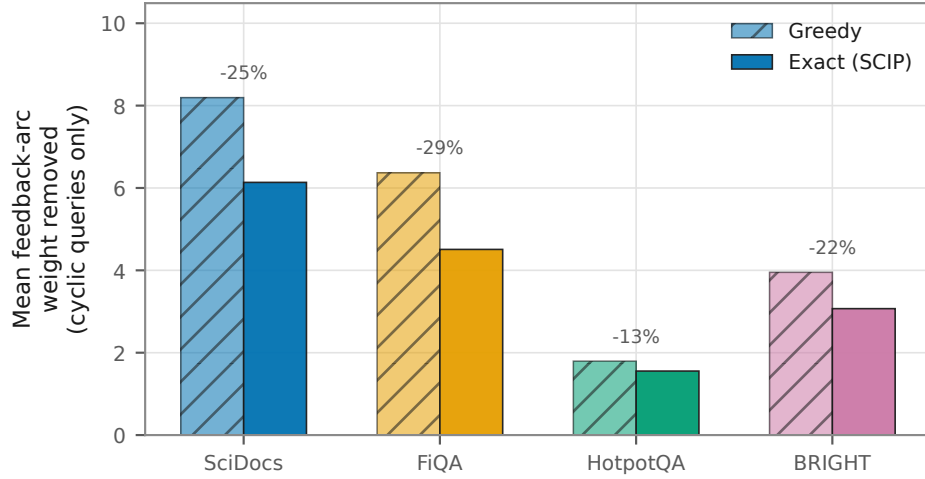
Fig. 4 Repaired-minus-unrepaired mean Δ nDCG with paired 95% bootstrap intervals, active **ms1** regime, primary five-pair method family. Point estimates vary in sign across datasets and methods, and nearly every interval includes zero.

what changes when repair is instead solved to certified optimality, addressing RQ4 directly: whether the retrieval null in Section 5.3 could instead reflect greedy repair’s suboptimality.

Of the 1,026 nominal query–regime combinations (342 usable queries $\times$ three vote-construction regimes), one BRIGHT query under **ms2** (**biology:0**) produces an empty edge set – no directed pair meets **ms2**’s min-support requirement of two rankers – and is excluded from the exact-repair study, leaving 1,025 valid exact-repair instances. Every one of those solves reached proven optimality: 1,025/1,025 query graphs, relative optimality gap 0.0, with a mean solve time of 7.4ms (median 0.25ms, maximum 236ms) against a 300-second per-query safety limit that was never approached (Section 4.6). Restricted to the 379 queries (pooled across all three vote-construction regimes) whose raw graph is actually cyclic, exact repair finds a different feedback arc set than greedy in 87.9% of them, and removes systematically less total edge weight: Table 7 and Figure 5 report the per-dataset gap. Greedy cycle-peeling is therefore a real, non-trivial overestimate of the minimum weighted feedback arc set on this evidence, not merely a theoretical possibility.

Table 7 Exact versus greedy repair, structural comparison, cyclic queries only ($n = 379$, pooled across vote-construction regimes). Mean weight removed is reported for both methods; the last column is the relative reduction from greedy to exact. Retrieval-level exact-versus-greedy and exact-versus-unrepaired Holm outcomes are reported in the accompanying text.

Dataset	Wt. removed (exact)	Wt. removed (greedy)	Reduction
SciDocs	6.14	8.19	25.1%
FiQA	4.51	6.37	29.2%
HotpotQA	1.55	1.80	13.4%
BRIGHT	3.07	3.95	22.3%



Exact repair consistently removes less total weight than greedy on the same graphs; both reach the same retrieval-level conclusion (no repaired-vs-unrepaired nDCG gain survives Holm correction under either repair method).

Fig. 5 Mean feedback-arc weight removed by greedy versus exact repair, cyclic queries only, by dataset. Exact repair removes less weight than greedy in every dataset, confirming that greedy over-removes rather than under-removes edge weight relative to the true optimum.

That structural gap does not translate into a retrieval-level difference. Pooled across all seven repaired-graph-dependent extraction rules evaluated in this comparison (Copeland, balance, Markov, both hybrids, topological, and priority-topological orderings; Section 4.7) and all five validated metrics, none of the 35 pooled metric-method cells is Holm-significant; at the finer 399-cell per-dataset-and-regime granularity, none is Holm-significant either. The largest raw, uncorrected effect in the pooled

Table 8 Robustness checks against the null retrieval finding (Section 5.3), canonical evidence only.

Check	Result
Baseline fairness	Testing whether the RRF-centered primary candidate pool gives a structural advantage to RRF-family baselines: 0/8 Holm-significant cells (RRF and CombSUM vs. the fixed repaired Copeland hybrid, canonical pool, all four datasets). No evidence of a pool-driven advantage.
Power and minimum detectable effect	In the active larger-pool family, median observed absolute mean delta 0.0036 vs. median Holm-adjusted 80%-power minimum detectable effect 0.0201: the study is well positioned to rule out larger effects, but many smaller ones remain below the corrected detectable scale.
Narrow equivalence	Two-one-sided-tests equivalence, active larger-pool family: 13/110 cells reject non-equivalence at a ± 0.005 margin, 32/110 at ± 0.010 , after Holm correction. Equivalence is established for a minority of cells, not the majority; the correct reading of the null result is absence of reliable positive evidence, not blanket practical equivalence.
Larger-pool family ($P > k$)	Already reported in Table 6: 0/110 Holm-significant despite a 10.6% mean top- k membership-change rate (Section 5.2), so membership change alone is not sufficient to produce a corrected retrieval gain.

family is a mean nDCG@10 difference of -0.00223 for priority-topological ordering (raw sign-flip $p = 0.026$; Holm-adjusted $p = 0.906$).

Exact repair also does not change the headline comparison against the unrepaired graph. Using the expanded nine-method family that adds PageRank, Rank Centrality, a Markov hybrid, and Bradley–Terry to the primary five-pair family specifically for this comparison (Table 4), 0 of 36 canonical exact-repaired-versus-unrepaired cells and 0 of 56 larger-pool cells are Holm-significant – the same qualitative outcome as the greedy-versus-unrepaired comparison in Section 5.3, now confirmed under a repair method that cannot be outperformed on the repair objective itself by any other feasible solution. Because exact repair improves the graph objective (Equation (3)) more than greedy repair does, yet does not reveal a retrieval gain of the scale detectable under this protocol, greedy under-repair is ruled out as an explanation for a retrieval gain of that scale in Section 5.3.

5.5 Robustness

Table 8 collects four canonical robustness checks against the primary retrieval finding, each drawn from a prespecified comparison family rather than a post-hoc search.

Each check addresses a distinct way the primary null result could have been an artifact of one specific design choice – an advantageous candidate pool, insufficient statistical power, or a $P = k$ evaluation that structurally hides repair’s effect – and none overturns it. The power analysis is the one qualification worth stating precisely: the exact-repair analysis rules out greedy under-repair as an explanation for a retrieval gain of the scale detectable under the evaluated protocol (median absolute delta 0.0036

versus Holm-adjusted median 80%-power MDE 0.0201), but a considerably smaller general effect, below that detectable scale, is not ruled out by this sample size.

5.6 Supporting LLM Evidence

A bounded pilot supplements the score-derived evidence above with real large-language-model pairwise judgments: six queries, four providers, and four preference-graph construction variants, analyzed at the query-cluster level ($n = 6$ independent clusters), since the pilot’s replicated rows within a query are repeated measurements, not additional independent samples. Whole-graph repair’s effect on retrieval, estimated the same way as the score-derived comparisons above, is a mean nDCG change of 0.00019 with a cluster-level 95% interval of $[-0.00072, 0.00140]$ (exact sign-flip $p = 0.875$): indistinguishable from zero, directionally consistent with the null finding in Section 5.3. Six clusters give this pilot very limited power, so this result is reported as directional and supporting only – it does not, on its own, extend any conclusion in this paper beyond the four benchmarks and construction regimes making up the main evidence package (Section 4.3).

6 Discussion

6.1 Main Findings

Taken together, the results answer the research questions with one restrained conclusion: under the canonical protocol, preference-graph repair is a real structural intervention – including under exact optimization – but not a validated general means of improving retrieval effectiveness (Sections 5.1–5.4). Structural consistency and downstream retrieval utility behave here as empirically distinct objectives, not as the same quantity observed at different resolutions.

That claim is narrower than it may first appear. It does not say that graph repair never helps retrieval, that acyclicity is useless, that exact repair always fails to improve rankings, or that the null of no effect has been proven. A null under corrected inference is evidence against a general, reliably detectable effect at the scale this study is powered to observe (Section 5.5); small or dataset-specific effects below that scale remain possible. On the evidence gathered here, treating acyclicity as a surrogate for retrieval effectiveness is not supported.

6.2 Why Structural Improvement Need Not Improve Retrieval

Section 3.5 already states the formal reason a repair-objective improvement need not track nDCG: the two objectives are computed from different information. This subsection interprets that gap in light of the results, distinguishing what the experiments establish from mechanisms consistent with – but not directly tested by – the design.

Several mechanisms are consistent with genuine structural repair producing no detectable retrieval effect, without any one of them being individually isolated. Feedback-arc-set repair targets the least-weight edges needed to restore acyclicity (Section 3.4); those need not be the edges that separate highly relevant from marginally relevant documents near the top of a ranking. A cycle broken far from

top-ranked positions can leave the extracted ordering of the most relevant documents unchanged. Where a hybrid mixes a graph-derived score with a prior (Equation (5)), an informative prior can also compress the influence of the graph-side score – consistent with graph-only and hybrid methods not diverging sharply in Section 5.3. Different extraction rules translate the same repaired graph differently (Section 4.7), which is one plausible contributor to the mixed-sign point estimates in Figure 4. Finally, pairwise preference weights and relevance judgments are built from different evidence (Sections 3.2, 3.1), so internal consistency and agreement with qrels are free to vary independently.

None of these mechanisms is demonstrated as *the* cause of the null retrieval result; they are offered as accounts consistent with the design and the observed pattern, not as claims established by a dedicated causal test.

6.3 Exact Repair as a Diagnostic Control

A heuristic-only repair study leaves open the alternative that a null retrieval result reflects under-repair rather than repair itself. Section 5.4 closes that alternative for this evidence: exact repair removes less edge weight than greedy yet still yields no Holm-surviving retrieval gain against the unrepaired graph. Exact SCIP repair is therefore used here as methodological control (Section 3.4), not as a novel solver, a novel formulation, or a recommendation that exact MWFAS be run at production scale (Section 2.3).

6.4 Relation to Prior Literature

The finding that graph repair is structurally active but not reliably retrieval-improving sits against three adjacent, and non-contradictory, strands of prior evidence. First, pairwise comparison signals genuinely improve ranking quality in several settings: pairwise ranking prompting with large language models is competitive with, and in some settings better than, pointwise scoring for text reranking [42], and pointwise, pairwise, and listwise LLM reranking paradigms show real, measured effectiveness differences depending on formulation [48]. Those findings are about whether *using* pairwise evidence, instead of pointwise scores, changes ranking effectiveness – a different question from this paper’s, which holds the pairwise evidence source fixed and asks whether *repairing* contradictions within that evidence changes effectiveness further. Nothing in this paper’s results is in tension with pairwise comparison being an effective ranking signal; the null result is specifically about the marginal effect of consistency repair on top of an already-constructed pairwise-derived graph.

Second, prior work documents that pairwise judgments, including LLM-generated ones, are prone to inconsistency and order sensitivity: positional and prompt-order biases can affect LLM-as-judge outcomes [61], systematic position bias has been measured directly in LLM-as-judge pairwise comparisons [46], and intrinsic inconsistency in LLM-based pairwise rankings has motivated explicit aggregation methods to mitigate it [59]. This literature establishes that inconsistency is a real, measurable property of pairwise preference data, which is the premise this paper’s construction-sensitivity results (Section 5.1) are consistent with. It does not by itself establish that *removing*

that inconsistency improves a downstream task metric; that is the further, separate claim this paper tests directly.

Third, and closest to this paper’s specific question, recent work on acyclic preference evaluation for language models treats acyclicity as a desirable property of an aggregated evaluation and reports downstream gains from enforcing it, on tasks including model ranking, response selection, and fine-tuning data selection [21]. That result is not contradicted by this paper: the two studies test acyclicity’s downstream value in different task settings (LLM evaluation and data selection versus document retrieval) with different aggregation and repair mechanisms, so a positive effect in one setting and an unsupported general effect in another are both consistent with the same underlying fact – that whether structural consistency helps a downstream task is an empirical property of the specific task and pipeline, not a property of acyclicity itself that transfers automatically across settings. The controlled separation used here for retrieval is stated in Section 2.4.

Relative to the earlier preprint discussed in Section 2.3, the present study is retrieval-focused and more conservative: it adds exact repair as a diagnostic control and applies multiplicity-corrected paired inference throughout. The two should not be read as successive measurements of the same effect; the preprint’s specific numerical findings are superseded by this protocol, not confirmed by it.

6.5 Practical Implications

The evidence in this paper supports a small number of restrained, directly evidence-backed recommendations for researchers and practitioners building preference-graph pipelines.

- Do not assume an acyclic or less-cyclic preference graph is a better retrieval input without checking; this study’s evidence does not support that assumption as a general rule.
- Evaluate repaired and unrepaired variants on the downstream task metric directly, with an unrepaired baseline always included as a comparison point, rather than reporting only structural diagnostics (cyclicity, removed weight) as if they were evidence of downstream benefit.
- Where a heuristic repair method is used and a null or weak downstream effect is observed, an exact solve on a representative subset of instances is an effective and specific check on whether heuristic suboptimality, rather than repair itself, could explain the result – this is a diagnostic use of exact solving, not a claim that exact repair must be run at full scale.
- Report graph-construction choices (normalization, vote semantics, candidate pooling), extraction rule, and evaluation cutoff explicitly alongside any repair claim: Section 5.1 shows these choices, not the repair algorithm, are what determine whether a graph has any cyclic structure for repair to act on in the first place.
- Treat structural-quality metrics and downstream-utility metrics as separate reporting dimensions with separate evidence, not as interchangeable indicators of pipeline quality.

These recommendations are audit practices, not a validated deployment rule for deciding when to apply repair. This paper does not establish, and does not attempt to establish, a criterion for predicting per-query whether repair will help; that is a distinct research question this paper’s evidence base does not address.

6.6 Implications for LLM-Based Ranking

Pairwise judgments from large language models can produce cyclic preference graphs for the same structural reasons as score-derived votes (Section 3.2). The distinction between structural consistency and downstream utility therefore applies to that setting in principle, but this paper’s own LLM evidence is only the bounded six-query pilot in Section 5.6 and does not support broad claims about modern LLM rerankers or production-scale pairwise pipelines. Extending the controlled comparison to a larger real-LLM evidence base remains future work.

7 Limitations and Threats to Validity

7.1 Internal Validity

The pipeline (Section 4.2) makes several sequential design choices – score normalization, vote-margin and aggregate-weight thresholds, candidate pooling, and the hybrid mixing weight α – each of which could in principle interact with the repair effect being measured. Two controls mitigate this: the retention-matched thresholding procedure (Section 4.5) ties the normalized protocol’s permissiveness to the raw-margin baseline rather than choosing it freely, and every retrieval comparison holds construction, pooling, and extraction fixed while varying only the repair method (Section 5.3), so a measured repaired-versus-unrepaired difference cannot be attributed to a confound in those other choices. What remains unresolved is α -dependence beyond the sweep reported as a robustness check (Section 4.7): the primary analysis fixes $\alpha = 0.3$, and while the sweep does not change the qualitative conclusion, it does not exhaustively rule out every possible mixing weight. The paired sign-flip and Holm-correction procedure (Section 4.10) assumes exchangeability of paired differences under the null and independence across queries within a family; the query-level pairing is designed to satisfy the first assumption, and queries are drawn from disjoint benchmark query sets to support the second, but formal independence across the three vote-construction regimes evaluated on the same underlying documents (Section 3.2) is not separately tested, since the canonical and larger-pool families are never jointly corrected across regimes for exactly this reason (Section 4.10).

7.2 Construct Validity

Acyclicity is one specific, narrow notion of structural consistency (Section 3.3); it is not the only property a preference graph could be judged by, and this study’s structural diagnostics (cyclic-query rate, removed feedback-arc weight, top- k membership-change rate) are proxies for graph-internal coherence, not direct measurements of evidential correctness. nDCG is this study’s primary utility construct; it captures rank-weighted agreement with graded relevance judgments, but it is one operationalization of

retrieval quality among several (Section 4.9 reports MRR and MAP only as secondary diagnostics, precisely because the primary claim should not rest on whichever metric happens to be most favorable). A further construct gap, not resolved by this study’s design, is that pairwise vote weight (Section 3.2) encodes a ranker’s confidence in a relative ordering, not a judgment of either candidate’s relevance; the two constructs are related in that both derive from the same underlying documents, but they are not the same measurement, and this study does not claim the graph’s internal weights are a proxy for relevance at any point (Section 3.5’s closing paragraph states this distinction as a matter of definition, not as an empirical finding requiring separate validation here).

7.3 External Validity

The main evidence base spans four benchmarks, two lexical base rankers, one dense base ranker, and three vote-construction regimes (Sections 4.3, 4.4) – a controlled, not exhaustive, sweep. It does not by itself establish generalization beyond that set. The real-LLM pilot remains the six-query addendum of Section 5.6 and is not an independent confirmation. This study does not evaluate learned fusion, cross-encoder reranking, modern dense rerankers beyond the MiniLM checkpoint used here, or online interactive retrieval; whether structural repair behaves differently under those settings is unaddressed. Extension to related preference-aggregation domains (recommendation, pairwise LLM evaluation, crowdsourced preferences) is plausible in principle (Section 6.6) but untested.

7.4 Statistical Conclusion Validity

Every confirmatory claim uses Holm-corrected paired inference over a reported family (Section 4.10). As reported in Section 5.5, the study is well positioned to rule out a general effect at the scale of the observed point estimates, but smaller effects below the corrected minimum-detectable scale are not ruled out; a non-significant result does not establish equality between repaired and unrepaired rankings, and narrow equivalence was established only for a minority of cells. The real-LLM pilot uses cluster-level ($n = 6$) inference for the reason stated in Section 4.10; that correction does not enlarge the pilot into a confirmatory study.

7.5 Computational Limitations

Exact MWFAS solving reaches certified optimality quickly at the query-graph sizes studied here (Section 5.4), but this study does not evaluate larger graphs and makes no scalability claim beyond those sizes. Exact repair’s role remains the diagnostic control of Section 6.3. Greedy cycle-peeling remains the practical choice for larger graphs because exact MWFAS is NP-hard in general (Section 2.3).

8 Conclusion

By separating graph construction, structural repair, ranking extraction, and retrieval evaluation, this study shows that construction choices decide whether preference

graphs are cyclic at all; that greedy and exact repair both restore acyclicity where structure exists, with exact repair removing less weight than greedy; and that neither repair method yields a Holm-surviving repaired-versus-unrepaired nDCG gain in the canonical, larger-pool, or exact-repair families – even when larger-pool repair changes top- k membership. Exact repair closes greedy under-repair as an explanation for a gain of the detectable scale without revealing a hidden retrieval gain; equivalence evidence remains limited to a minority of TOST cells, and smaller effects below the corrected minimum-detectable scale remain possible.

The practical implication is not that acyclicity is irrelevant or that repair can never help. It is that structural improvement and retrieval improvement are different claims requiring different evidence: report construction, repair diagnostics, extraction, and downstream metrics separately, keep an unrepaired baseline, and use exact repair as a bounded methodological control when a weak downstream result must be stress-tested (Section 6.5).

Future work should test the same separation on broader large-language-model judgment benchmarks, additional construction and extraction strategies, and downstream objectives beyond nDCG-based retrieval, while preserving the discipline established here: evaluate acyclicity as a structural property first, and as a retrieval intervention only when downstream evidence supports that reading.

9 Data Availability and Reproducibility

The public benchmarks used in this study are available from their original sources: SciDocs, FiQA, HotpotQA, and BRIGHT are cited in Section 4.3. The code, fixed query lists, processed intermediates, figure data, and scripts required to reproduce the reported tables and figures are available at <https://github.com/SoroushVahidi/consistency-aware-llm-rankin>. The main-paper score-derived results are regenerated from stored ranker-score and relevance-judgment artifacts; no paid provider call is required to recompute those statistics. The bounded real-large-language-model pilot reported in Section 5.6 is supported by sanitized derived artifacts and usage summaries in the repository. Raw provider request and response payloads are not included in the repository, because they may contain prompt/completion content and operational details that are excluded by the project’s artifact policy.

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Statements and Declarations

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Competing interests

The author declares no competing interests.

Ethics approval

Not applicable. This study uses only publicly available benchmark datasets and stored, derived score files; it involves no human subjects, no personal or identifiable data, and no new data collection from human participants.

Consent to participate

Not applicable; no human participants were involved in this study.

Consent for publication

Not applicable; no individual person’s data are presented in this manuscript.

Data, materials, and code availability

See Section 9 for dataset sources, repository location, and the exclusion of raw provider payloads from the bounded large-language-model pilot.

Authors’ contributions

This is single-author work. Soroush Vahidi conceived the study, designed and implemented the experimental pipeline, ran the experiments, analyzed and verified the results, prepared the figures and tables, and wrote the manuscript.

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